

## **ADS508 Data Science With Cloud Computing**

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**Company Name:** J&F Realty Insighters

**Company Industry:** Real Estate Analytics

**Company Size:** 2

### **Abstract:**

J&F Realty Insighters is a real estate analytics company that provides data driven insights into housing market trends. Currently facing the challenge of accurately identifying the factors that influence housing price trends, this project will develop a machine learning model that predicts housing price trends by analyzing housing supply, demand indicators, and historical home values.

### **Problem Statement:**

Real estate investment has traditionally relied on historical trends, expert intuition, and manual data analysis. However, the housing market is a high-stakes environment shaped by a complex web of variables—from hyper-local demographics to broad economic shifts. Without a rigorous, data-driven framework, investors risk inaccurate valuations and missed opportunities in emerging markets.

J&F Realty Insighters bridges this gap. By utilizing machine learning to synthesize massive housing and demographic datasets, we provide precise property value estimations and identify the hidden patterns driving market growth. Our predictive models empower investors to make informed, high-conviction decisions while significantly mitigating financial risk.

### **Goals:**

- **Predict:** Deliver precision property valuations through advanced machine learning.
- **Analyze:** Pinpoint the key demographic and economic factors driving home prices.
- **Scale:** Leverage AWS to build a robust, cloud-native data architecture.
- **Identify:** Help investors discover the next high-growth real estate markets.
- **Optimize:** Prove that data-driven insights outperform manual analysis in property investment.

### **Non-Goals:**

- **Financial Advice:** This project provides data analysis, not personalized investment or financial counseling.
- **Real-Time Data:** Predictions are based on historical datasets rather than live, real-time market feeds.

- Production Deployment: We will not be integrating the model into a commercial-grade real estate platform.
- Proprietary Access: Integration with private or third-party proprietary databases is excluded.
- Short-Term Volatility: Our focus is on long-term trends rather than daily or weekly price fluctuations.

**Data Sources [<https://www.zillow.com/research/data/>]:**

- Zillow Home Value Index (ZHVI): Represents the estimated typical home value and serves as the primary target variable for our predictive model.
- Median Sale Price: Provides historical transaction data to validate model accuracy against actual market sales.
- Market Temperature Index: Measures supply-demand competitiveness to identify "hot" seller's markets.
- Housing Inventory: Tracks available for-sale stock to quantify the supply-side impact on price fluctuations.

**Technical Workflow**

The data architecture is built for scale and reproducibility using the AWS ecosystem:

- Storage: Raw historical time-series data is staged in Amazon S3.
- Ingestion: Data is pulled into AWS SageMaker Studio for exploratory analysis and development.
- Transformation: We convert datasets from wide to long format, merging them via shared geographic identifiers (Region/State) to create a unified feature set.
- Modeling: This integrated dataset powers a regression model designed to forecast price trends based on supply and demand dynamics.